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Defination

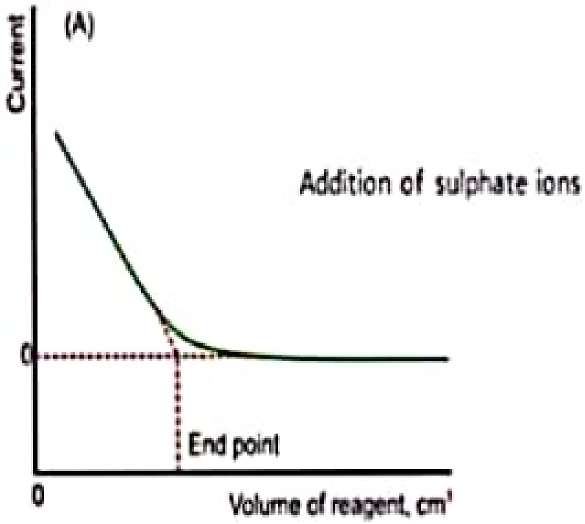
- Amperometric titration refers to a class of titration in which the equivalent point is determined through measurement of the electric current produced by the titration reaction. It is a form quantitative analysis.
- Other wise called as polarographic or polarometric titration
- Polarography can be used as the basis of amperometric titration method comparable with the potentiometric , the conductometric , and the photometric methods.
- From polarographic technique , it is known that current is independent of applied voltage impressed upon dropping mercury electrode or any other electrode .
- In amperometric titration the voltage applied across the indicators electrode and the reference electrode is kept constant and diffusion current passing through the cell is measured and plotted against the volume of the reagent added .
- The fundamental basis of amperometric titration derived form ampere and unit of current.

Principle

- The principle is that ,the potential applied between polarizable and non-polarizable electrode is kept constant and the diffusion current is measured during the titration.
- During the titration the concentration of electo reducible ion change and hence the diffusion current also change.
- At the principle is that the potential applied between polarizable and non-polarizable electrode is kept constant and the diffusion current is measurement during the titration.
- The end point there is sharp change in the diffusion current as shown by the curve of diffusion current v/s volume of titrant.
- The titration is perform between a reducible or non-reducible ion and a counter ion of which at least the titrate or the titrant or both arise to diffusion current.
- The potential selected for the titration is at its limiting value that is the potential compounds to the point where limiting current.

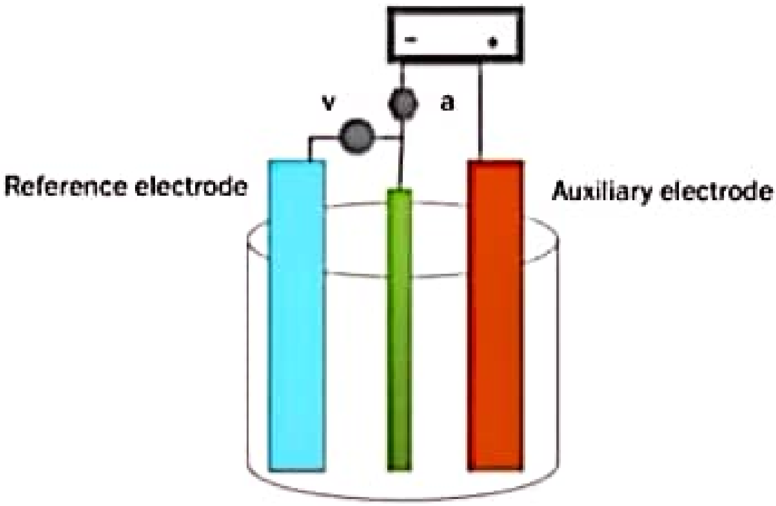
Example :

Precipitation of a electro reducible ion gradually causes as a decreses in diffusion current.



- The first part of curved shows a decrease in current due to decrease in the concentration or lead ion in solutions.
- This is due to precipitation as lead sulphate by sulphate ions {titrant}.
- At the end point as all the lead ions are removed the diffusion current value reaches a minimum.
- The addition of sulphate ions after the end point dose not cause any change in diffusion currents as it an non-reducible.
- The intersection of two lines corresponds to the end point of such titration.
- The lead ions (titrate is electro reducible) is titrated is v/s sulphate ion (titrant is non-reducible). Diffusion current is observed due to lead only.

Cells with three electrodes



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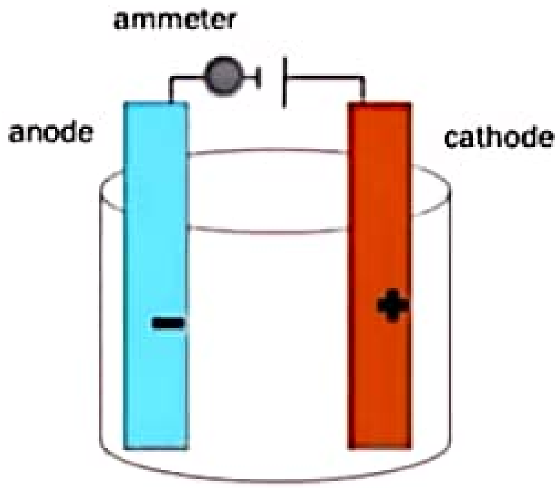


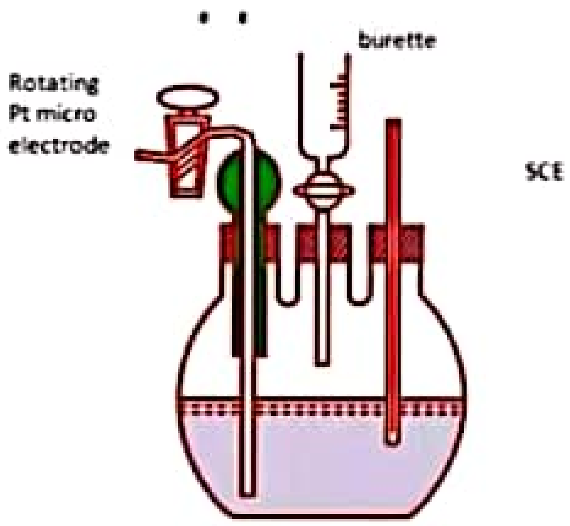
CONDITION FOR PERFORMING AMPEROMETRIC TITRATION

- Either the titrate or titrant or both should be electro reducible.
- The potential applied should correspond to the limiting current.

Apparatus used for amperometric titration

- The equipment used in the case of amperometric titration is simple although it ,may be same as used for polarography several simplification are possible.
- The titration may be performed either with the dropping mercury electrode or with the dropping mercury cell modified to allow the titrating reagent.
- Calomel electrode is used as a reference electrode.
- The galvanometer measure the current and the series of rheostat may be used for changing the sensitivity of the galvanometer.
- It is the voltage applied to the indicator electrode and the reference electrode is kept constant.
- The sensitive galvanometer indicates the value of diffusion current after each addition of the titrant.
- The sensitive galvanometer generally the potential of a reference electrode will be lie in permissible range so that it is necessary only to short circuit. The indicator electrode through a suitable measuring instrument to reference electrode of relatively large area.

Apparatus



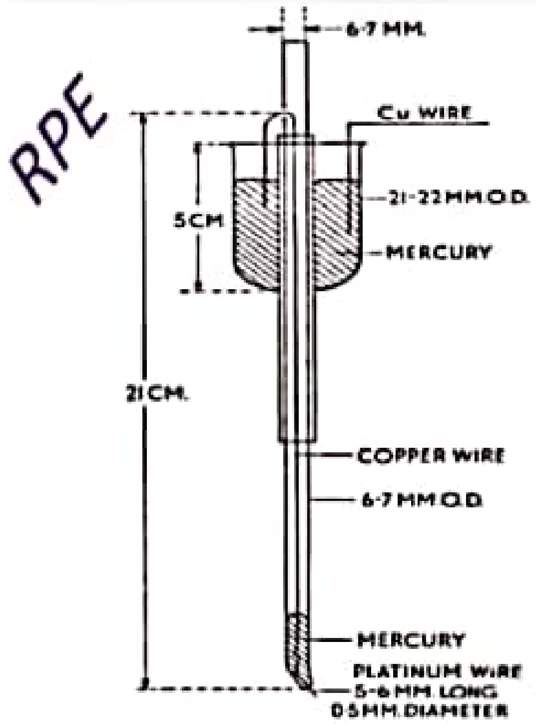
Apparatus

- Pyrex glass , that is three necked ,flat bottomed flask.
- A micro burette.
- Dropping mercury electrode (indicate polarizable).
- Gas outlet for N2 , with an additional inlet N2 provision.
- Connected to a reference electrode (Non-polarizable)

ROTATING PLATINUM MICRO-ELECTRODE

- The rotating platinum micro –electrode was introduced by H. A .Laitinen and I .M. Kolthoff in 1941.
- It consist of glass tube about 15-20 cm in length and 6mm in diameter.
- A short length of platinum wire extends 5-20 mm from the fall of the glass tube.
- The simple rotatory platinum electrode arrangement used in amperometric titration.
- In amperometric titration removal of oxygen is necessary if the electrolysis is carried out at an E.M.F, at which oxygen would have a diffusion current.
- Removal of oxygen is done by bubbling purified nitrogen before of the titration and for about 1mim.after each addition of the titrant.

RPE



Rotating platinum electrode

- **WHY platinum??**
- Because mercury can not be used as electrode at positive potential of its oxidation.
- rotating platinum is used.

Why rotating ??

- Because with platinum as electrode , the attainment of steady state diffusion current is slow.
- One has to wait for a considerable time after each addition of the reagent.
- Therefore platinum electrode is rotated at 600 rpm.

Advantage of RPME

- It is simple construct.
- It increases the workable range on the positive voltage side up to 0.9v Thus it can be used at positive potential where as the mercury electrode may be used.
- The technique is more sensitive because the rotation of the electrode increase the value of diffusion current as much as time the value in polarography.

AD

Dead stop point method OR Titration with two indicator electrode method :-

* This method is applicable only when the oxidation reduction system is involved before and after the end point .

* In this method two similar platinum electrode are immerised in the titration cell. Then small and constant voltage is applied to these two electrodes.

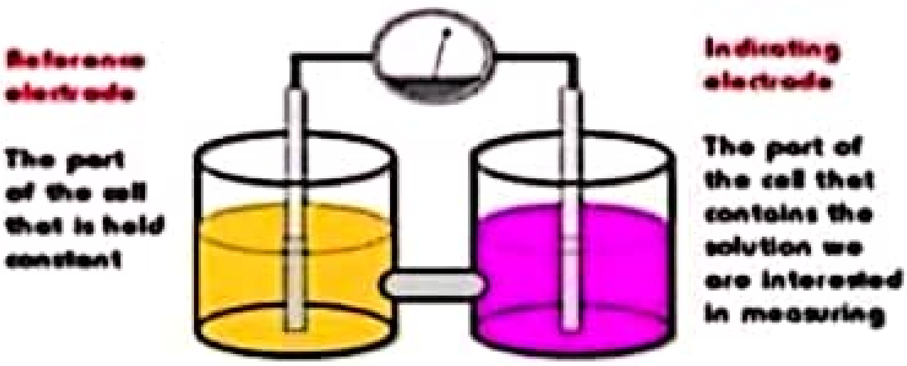
* An electrometric titration apparatus with the two electrodes dipping in two different vessels and linkage though a salt bridge.

* The amount of oxidized from reduced at the cathode is equal to the form by oxidation of the reduced from at the anode when the reactant involves a reversible system.

* At this stage with electrodes are depolarized until either the oxidized or the reduced has been consumed by the titrant.

* After the end point only electrode remains depolarized as if the titrant does not involve a reversible system.

Titration with two indicator electrode

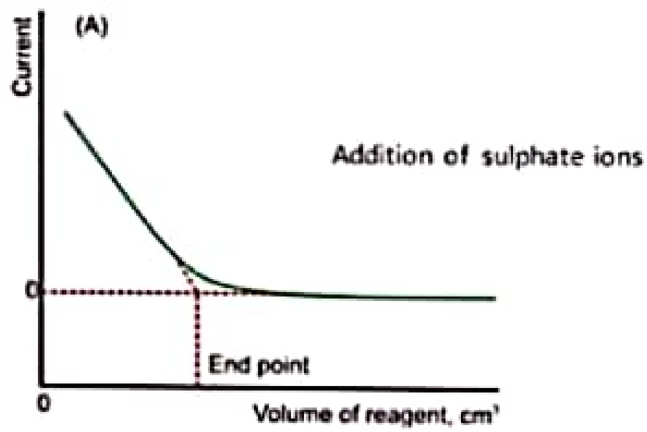


TYPES OF AMPEROMETRIC TITRATIONS

- Titration of non-reducible ion V/s electro reducible ion .
- Titration of electro reducible ion V/s electro reducible ion.
- Biampereometry
- Titration of electro reducible ion V/s non-reducible ion .

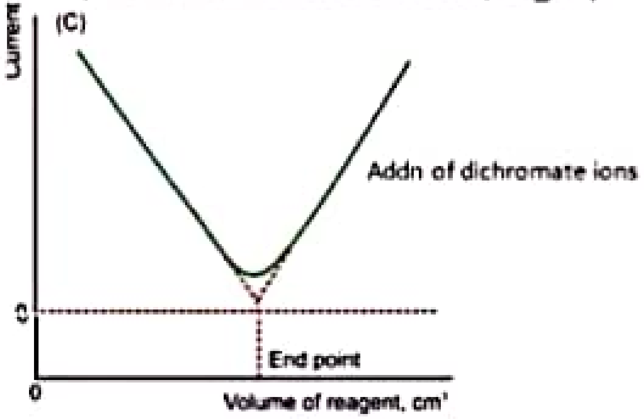
Titration of electro reducible ion v/s non-reducible ion

- Ex: lead(Pb^{2+})Vs sulphate ions (SO_4^{2-})



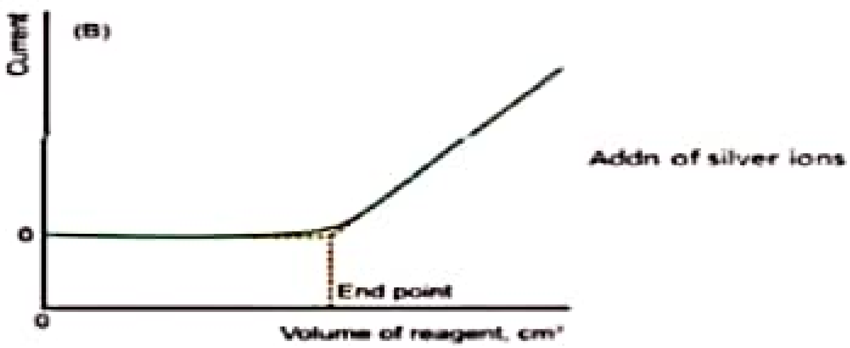
Titration of non-reducible ion v/s electro reducible ion

- Ex: lead (Pb^{2+}) Vs dichromate ions($Cr_2O_7^{2-}$)



Titration of electro reducible ion v/s electro reducible ion

- Ex: chloride(Cl^-) Vs silver ions(Ag^{2+})



ADVANTAGES OF AMPEROMETRY

- Rapid, simple apparatus
- Titration can be carried out in cases in which the solubility relations are such that potentiometric or visual indicator methods are unsatisfactory.
- Can be carried out at dilution (10-4M) at which many visual or potentiometric titrations no longer yield accurate results.
- These titration can be carried out rapidly because the end point is found graphically.
- It is not necessary to have capillary characterise of the dropping electrode.
- The reaction carried out can be reversible or irreversible.
- Dilute solution can be analysed.
- The method is relative therefore , fewer distributing factors are prevalent.

DISADVANTAGES OF AMPEROMETRY

- **In accurate results are sometimes obtained because of co-precipitation.**
- **The foreign substance which do not interfere in the amperometric titration should not be present in larger concentration than the substance to be filtrated.**